Multi-Task Self-Supervised Learning for Automated Measurement of Left Ventricular Ejection Fraction in Echocardiography

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Background & Motivation

Clinical Context: LVEF is a critical metric for assessing cardiac function and guiding treatment.

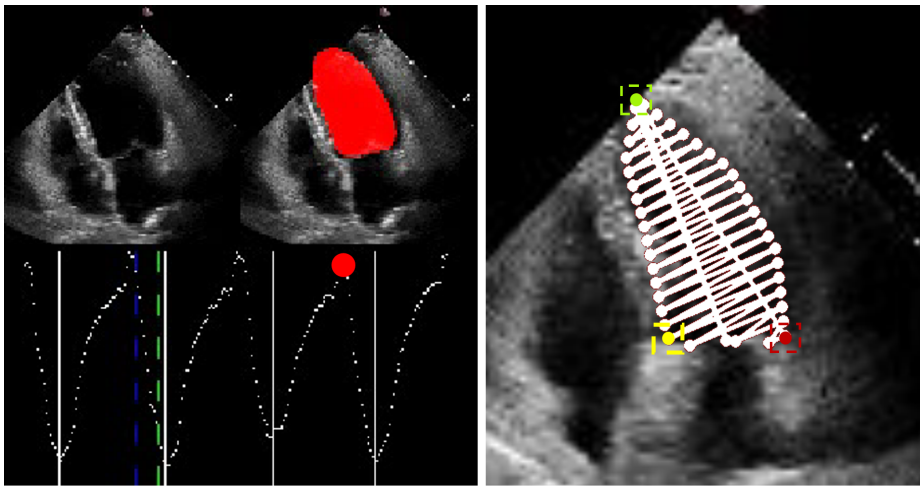


Figure: Manual echocardiographic analysis by clinicians.

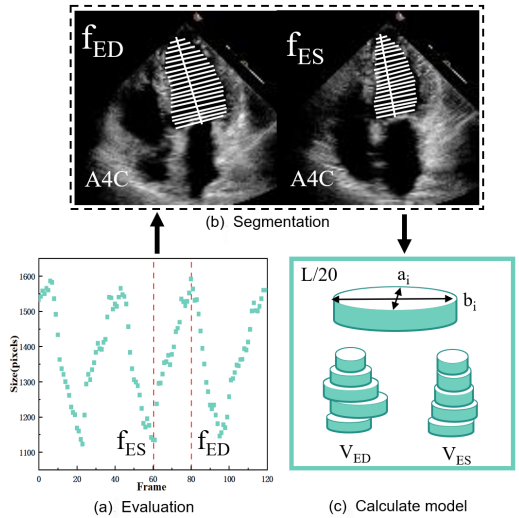


Figure: LVEF estimation process via Simpson's method.

Challenges:

- Manual annotation is time-consuming and inconsistent.
- Existing models often fail to locate key landmarks.

Objective: Design a lightweight, multi-task self-supervised framework for accurate and automated LVEF estimation under sparse annotation.

Overview of SSP-MAEFNet

- **Stage 1:** Self-supervised pretraining with random masking and frame reconstruction to learn spatio-temporal features.
- **Stage 2:** Weakly-supervised segmentation and landmark detection via dual-branch decoder with DSTS module.
- **Stage 3:** LVEF estimation using Simpson ‘s method based on segmentation and landmark results.

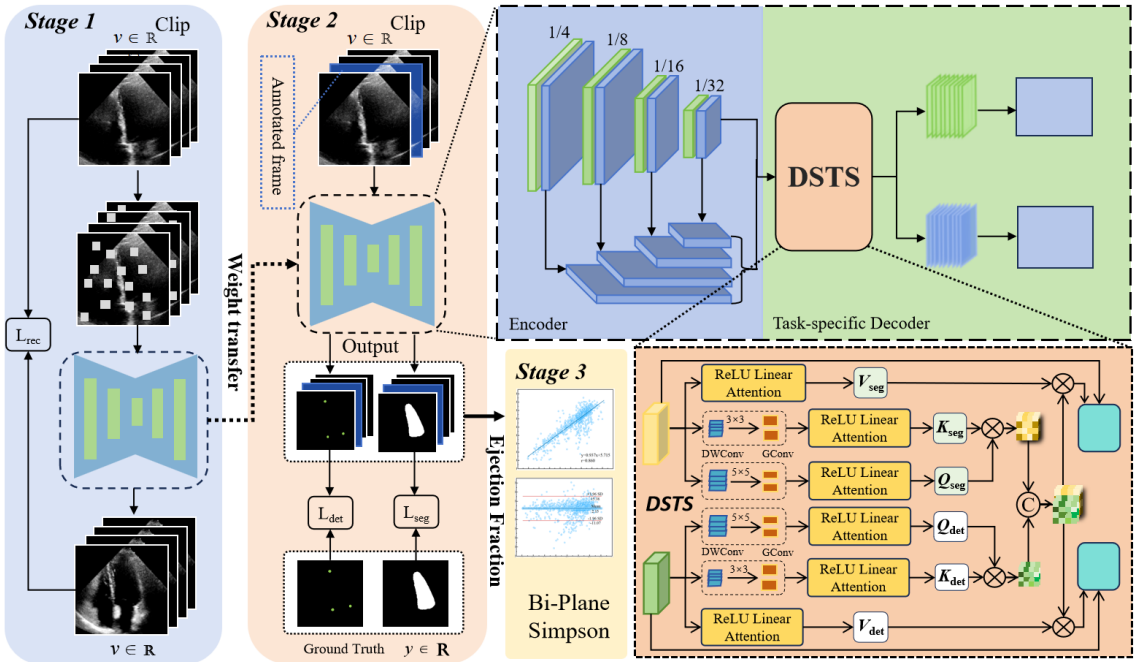


Figure: Overview of SSP-MAEFNet

Quantitative Results on EchoNet-Dynamic

Highlights:

- Achieves highest Dice (93.54%) and lowest HD (3.512) for segmentation with only 0.25G FLOPs.
- Outperforms all landmark detection baselines with OKS of 81.30 and smallest model size (5.0M).

Segmentation Results

Model	DSC↑	HD↓	Params	FLOPs
U-Net	92.48	4.501	14.7M	10.4G
nnU-Net	92.83	4.379	19.2M	14.3G
EchoGraph	91.62	4.685	27.6M	21.5G
MemSAM	93.15	4.028	138.9M	109.6G
EchoNet	92.23	4.253	39.6M	10.6G
Ours	**93.54**	**3.512**	5.0M	**0.25G**

Landmark Detection Results

Model	OKS↑	Params	FLOPs
HRNet	74.88	62.7M	3.7G
MSPN	76.39	27.5M	1.56G
AlphaPose	80.30	28.3M	3.51G
RTMPose	80.90	28.6M	1.06G
Ours	**81.30**	**5.0M**	**0.25G**

Visual Comparison of Segmentation and Landmark Detection

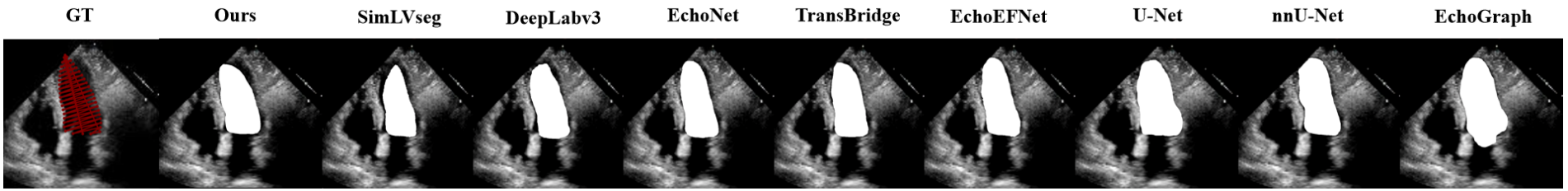


Figure: Segmentation comparison across models. Our method yields more accurate and sharper left ventricular boundaries.

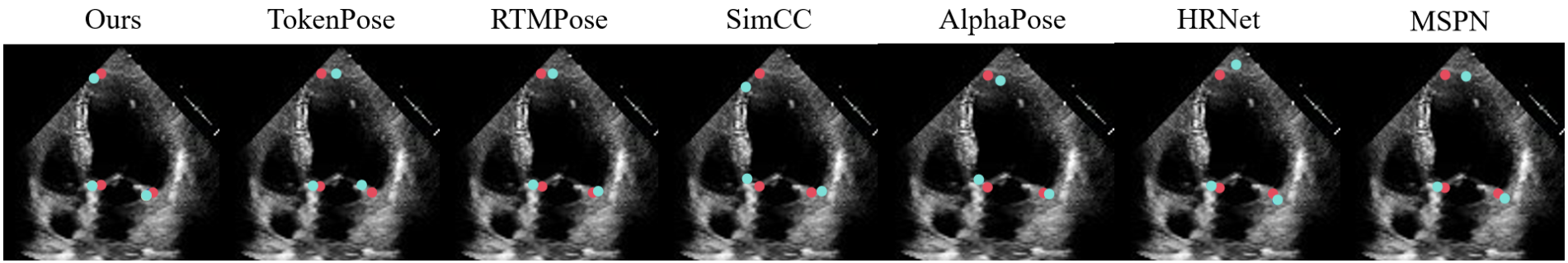


Figure: Landmark detection in challenging regions. Our model performs better in blurred areas, such as the lateral mitral annulus and apex.

Effectiveness of DSTS and Pretraining

Key Findings:

- **DSTS module** consistently improves performance across diverse backbones (Table 1).
- Combining **DSTS with self-supervised pretraining** further boosts accuracy with minimal overhead (Table 2).

Table 1: DSTS Across Different Backbones

Backbone	DSTS	Dice↑	HD↓	OKS↑
MobileNet	✓	80.33	4.902	0.754
ResNet50	✓	92.05	3.791	0.742
SwinTrans	✓	90.94	4.083	0.728
PoolFormer	✓	89.44	5.216	0.731
DeiT	✓	86.62	5.428	0.717
EffFormer	✓	91.10	**3.645**	0.764
ReoViT	✓	91.89	3.812	0.790
MobileNetV3	✓	**92.47**	4.024	**0.795**

Table 2: Combined Effects of DSTS + Pretraining

Model	Dice↑	HD↓	OKS↑	Params	GFLOPs
Base	91.56	4.139	0.778	4.95M	0.23
+ DSTS	92.47	4.024	0.795	5.01M	0.25
+ DSTS + Pretrain	**93.54**	**3.512**	**0.803**	5.01M	0.25

Evaluation of EDV, ESV, and LVEF Predictions

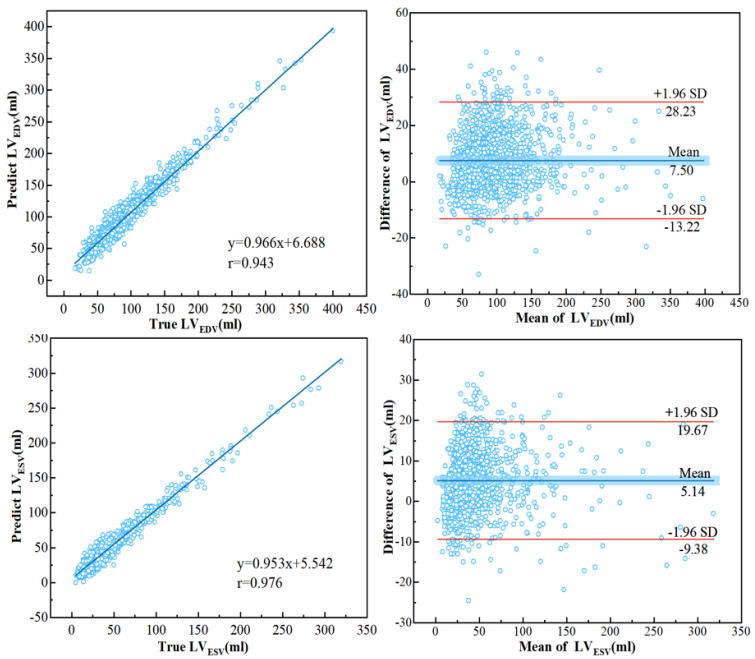


Figure: Correlation and Bland–Altman plots of EDV, ESV, and LVEF.

Index	Corr	LOA	PELOA	ICC
EDV	0.943	21.06	23.06	0.943
ESV	**0.976**	**14.85**	**34.07**	**0.954**
LVEF	0.860	13.45	23.19	0.787

- ESV shows strongest agreement with ground truth.
- LVEF correlation is weaker due to error accumulation.
- ICC = 0.787 indicates clinically acceptable reliability.